**Course:** MCA /MSc (IT)/MSc (IT) DS

**Semester: 3**

**Rationale:** The key objective of this course is to introduce the fundamentals of web development and build dynamic web applications using the Flask framework. It also aims to develop skills in database integration, user authentication, and deployment of scalable web applications.

**Prerequisite:** Basic knowledge of computers, HTML, and fundamental programming concepts such as variables, functions, and logic is recommended.

**Course Content**

| Sr. | Topics | Hrs |
|---|---|---|
| 1 | **Module 1: Introduction to Flask and Web Development Concepts :**<br>Fundamentals of Web Development, Flask vs Django overview, Introduction to Flask: Architecture and Advantages, Flask installation and setup using pip and virtual environments, Creating the first Flask app: "Hello World" ,Understanding Flask project directory structure.<br>**Lab Work**:Build and run your first Flask web application,Use app.route() for defining routes. | 4 |
| 2 | **Module 2: Routing, Templating and Static Files:**<br>Understanding URL Routing,Dynamic URL building using url_for(),Template engine: Jinja2,Template inheritance and filters,Passing data to HTML templates,Static files (CSS, JS, images).<br>**Lab Work**:Build multipage navigation,Create reusable base templates and use Bootstrap. | 4 |
| 3 | **Module 3: Forms and Request Handling :**<br>Working with HTML forms,HTTP methods: GET, POST,Handling form data with request object,File uploads and input validation<br>**Lab Work:**Create input forms with validations,Handle file uploads and form responses | 4 |
| 4 | **Module 4: Flask with Database Integration:**<br>Basics of Relational Databases (SQLite/MySQL),Introduction to SQLAlchemy ORM,Performing CRUD operations with Flask + SQLAlchemy,Migrations with Flask-Migrate,Database session management<br>**Lab Work**:Build a CRUD application (e.g., task manager or student records) | 4 |
| 5 | **Module 5: User Authentication and Sessions:**<br>User login/logout using Flask-Login,Managing sessions and cookies,Flash messaging,Creating registration and profile update forms,Role-based access control.<br>**Lab Work**:Build a user registration/login systemMaintain user session and restrict routes by login status | 4 |
| 6 | **RESTful API and Deployment:**<br>Introduction to RESTful APIs,Creating REST endpoints with Flask,JSON data exchange,Organizing project with Flask Blueprints,Error handling and logging,Deployment options (Heroku, Render, PythonAnywhere)<br>**Lab Work**:Build a simple API with FlaskCreate modular Flask apps using Blueprints | 4 |
| 7 | **Module 7: Project Work** | 6 |
| | **Total** | 30 |

**Reference Books**

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| 1. | **Flask Web Development – Miguel Grinberg**<br>**A comprehensive guide to building web applications with Flask.** |
| 2. | **Python Web Development with Flask – Samuli Natri**<br>**Covers Flask fundamentals and real-world applications** |
| 3. | **Learning Flask Framework – Matt Copperwaite**<br>**Beginner-friendly introduction to Flask development.** |
| 4. | **Flask By Example – Gareth Dwyer**<br>**Practical, project-based approach to learning Flask.** |

| **Course Outcome** |
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| **After Learning the Course the students shall be able to:** |
| 1. Understand web development fundamentals and build basic web applications using the Flask framework.<br>2. Design dynamic web pages using routing, templates (Jinja2), and handle user input through forms and HTTP requests.<br>3. Integrate databases using SQLAlchemy and perform CRUD operations for data-driven applications.<br>4. Develop secure and scalable web applications with authentication, RESTful APIs, and deploy them on cloud platforms. |